Predicting CO2 savings from renewables

Dominika Filipiak, Emilia Velikova, Danu Cislaru, Kacper Grobelny, Filip Bičanić

Why Do CO_2 Savings from Renewables Matter?

- Energy production is one of the largest sources of global CO_2 emissions
- Fossil fuels significantly contribute to climate change
- Renewable energy reduces emissions by replacing fossil fuel generation

Project Goal

- Predict CO emissions per kWh using machine learning
- Analyze how different energy sources affect emissions
- Identify which energy sources are the most impactful

Research Objective

- Quantify how much CO_2 emissions are avoided by renewable energy production
- Analyze the relationship between renewable share (%) and CO_2 savings
- Evaluate how increasing renewable energy impacts emission reduction
- Compare impact across countries and over time

Dataset overview

The global dataset covers all countries around the world

Time range: last 10 years
[2016-2025]

Contains energy production by source: coal, gas, oil, hydro, nuclear, solar, wind and biofuel

Includes renewable share (%) and carbon intensity (CO_2 per kWh)

Dataset overview and preprocessing

```python
print("Dataset information:")
print(df.info())

print("\nMissing values per column:")
print(df.isnull().sum())
```

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1758 entries, 0 to 1757
Data columns (total 12 columns):
 #   Column                                  Non-Null Count  Dtype
---  ------                                  --------------  -----
 0   country                                 1758 non-null   object
 1   year                                    1758 non-null   int64
 2   biofuel_share_elec                      1758 non-null   float64
 3   coal_share_elec                         1758 non-null   float64
 4   gas_share_elec                          1758 non-null   float64
 5   hydro_share_elec                        1758 non-null   float64
 6   nuclear_share_elec                      1758 non-null   float64
 7   oil_share_elec                          1758 non-null   float64
 8   other_renewables_share_elec_exc_biofuel 1758 non-null   float64
 9   solar_share_elec                        1758 non-null   float64
 10  wind_share_elec                         1758 non-null   float64
 11  carbon_intensity_elec                   1758 non-null   float64
dtypes: float64(10), int64(1), object(1)
memory usage: 164.9+ KB
None

Missing values per column:
country                                   0
year                                      0
biofuel_share_elec                        0
coal_share_elec                           0
gas_share_elec                            0
hydro_share_elec                          0
nuclear_share_elec                        0
oil_share_elec                            0
other_renewables_share_elec_exc_biofuel   0
solar_share_elec                          0
wind_share_elec                           0
carbon_intensity_elec                     0
dtype: int64
```

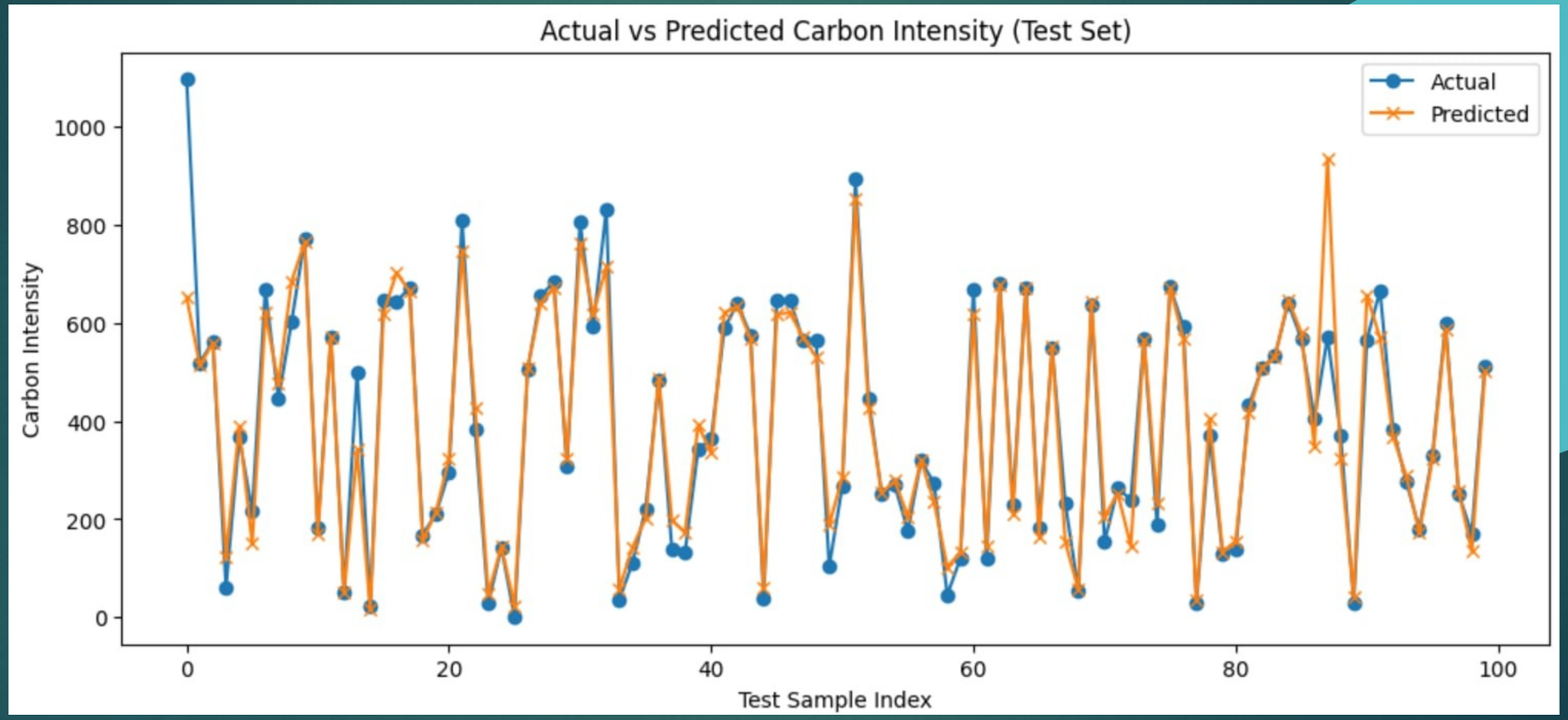
```python
# Forward + backward fill (ca în PDF)
df = df.ffill()
df = df.bfill()
df = df.dropna()

print("Missing values after cleaning:")
print(df.isnull().sum())
```

```
Missing values after cleaning:
country                                   0
year                                      0
biofuel_share_elec                        0
coal_share_elec                           0
gas_share_elec                            0
hydro_share_elec                          0
nuclear_share_elec                        0
oil_share_elec                            0
other_renewables_share_elec_exc_biofuel   0
solar_share_elec                          0
wind_share_elec                           0
carbon_intensity_elec                     0
dtype: int64
```

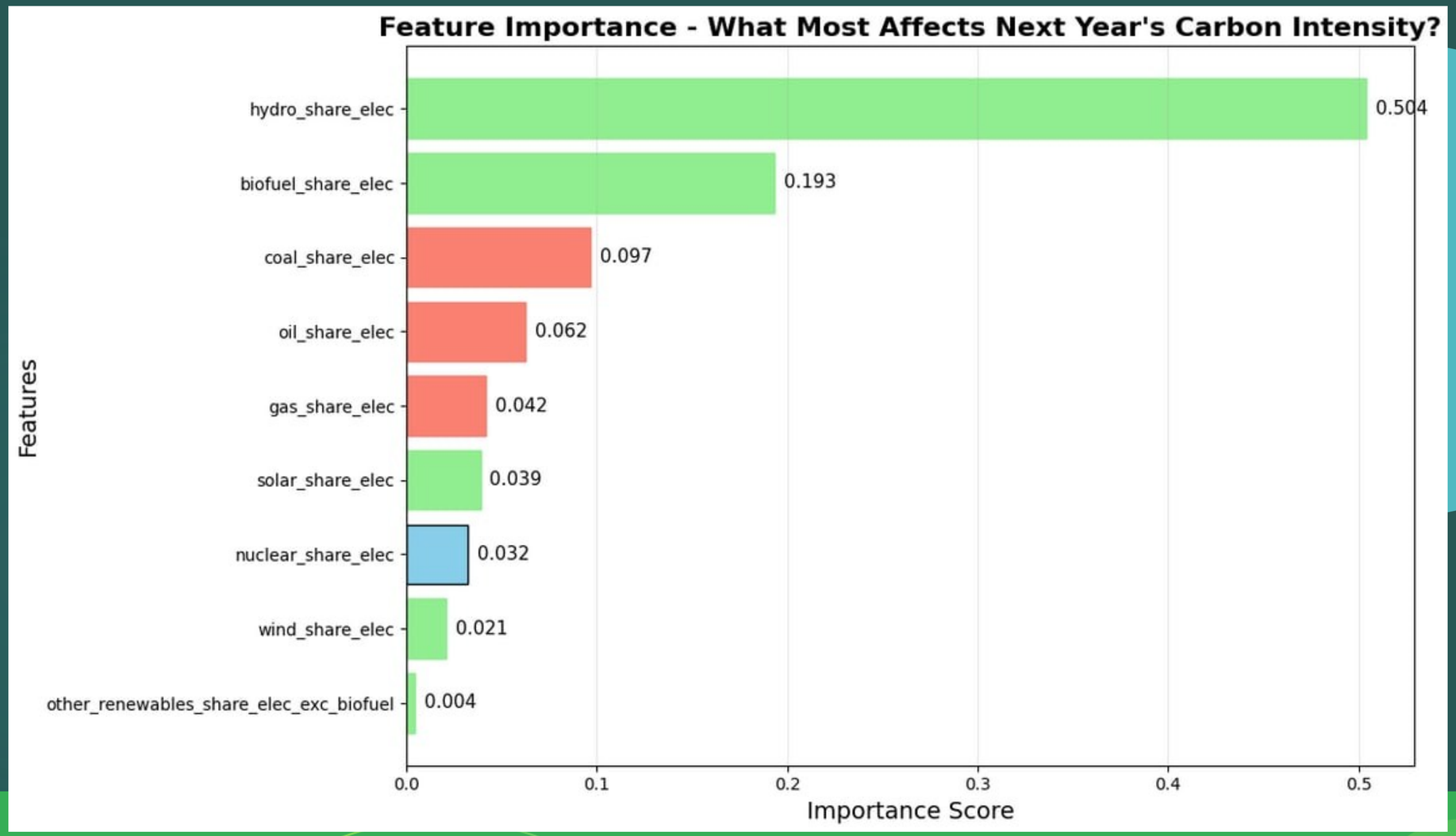
How We Approached The Problem

- Cleaned and prepared the dataset
- Created prediction target (next-year emissions)
- Split data into training and testing sets
- Trained a Random Forest model
- Evaluated performance using multiple metrics

Model Predictions:
Actual vs Predicted CO_2 Emissions



What Factors Influence CO_2 Emissions the Most?



THANK YOU FOR YOUR ATTENTION!